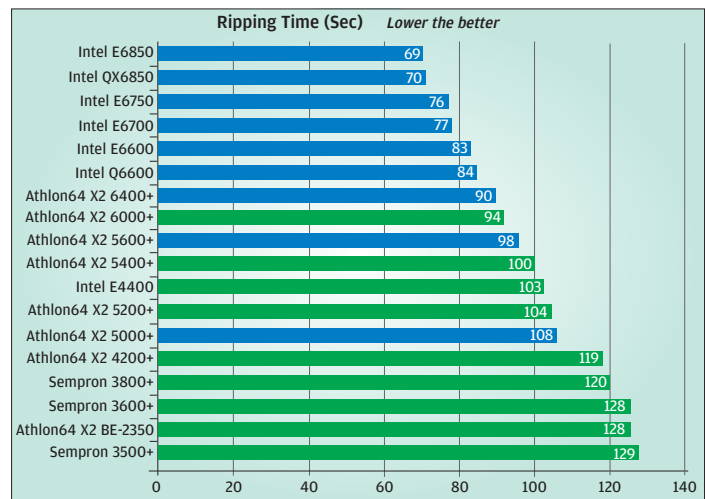
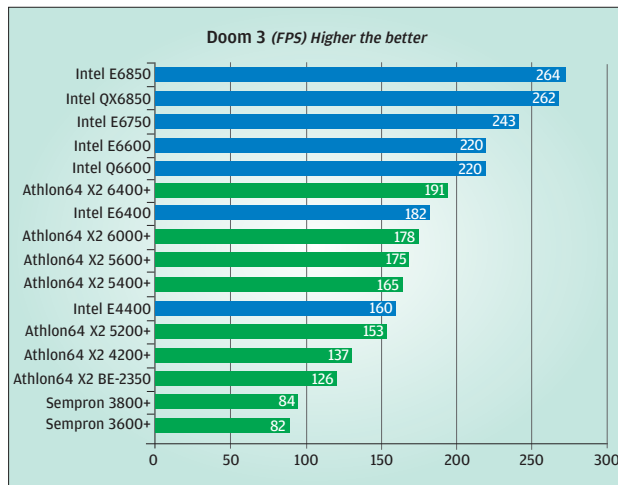


see negligible performance gains. Except for the E4400, all other Intel processors gave us better fps scores than AMD's top-end processors. The Athlon64 5600+ fell just shy of the E4400 by one fps, and we think it offers the best value for money among the AMD bunch.

Doom 3

The results for *Doom 3* are quite similar to other two games—the quad-core processors from Intel fall short of expectations.

Intel's processors are clear winners in the gaming arena; no gaming PC should be without one. Interestingly, AMD's Athlon64 X2 5200+ does come close to Intel's new E4400, and offers similar value.



DivX Encoding

The graph you see is more the shortcoming of the DivX encoder than the processors—the encoder hasn't incorporated code for multiple processors. We see better results with increasing clock speeds, and not multiple cores. Again, the quad-core processors trail their dual-core counterparts, albeit by a very small margin. All the top-end Intel processors outdo AMD by a significant margin. The duel between Athlon64 X2 5200+ and E4400 should be noted again—they're separated by a hair's breadth. The Athlon64 X2 BE 2350 and 2300 trail behind the Semprons—this again is because the encoder favours higher clock speeds, so the lower-clocked BEs take a hit.

How We Tested

Machine Configuration

All Intel LGA 775-based processors are tested on an ASUS P5K Deluxe motherboard, based on Intel's P35 Chipset. For AMD processors, we use an ASUS M2A-VM HDMI motherboard, based on AMD's 690 chipset. Corsair stays our choice for memory—we used 2 GB of 1066 MHz RAM at 4-4-4-12 settings (wherever possible). An XFX 8800 GTX graphics card was used on this rig, with NVIDIA's ForceWare version 158.22. On the software front, Windows XP with SP2 was freshly loaded on a Hitachi 7200 RPM, 250 GB SATA drive. CoolerMaster Real Power Pro 550 watt powered the rig. The drive was defragmented before each new processor run. The motherboards used were updated with the latest available BIOS patches; driver releases for chipsets and other critical peripherals were updated as well.

Benchmarks

PCMark 05 and 3DMark 06: These benchmarks from FutureMark are widely used to gauge performance on a system- as well as sub-system-level. Though mostly related to graphics performance, each of these benchmarks has a really well-developed CPU test that taxes the processor to the maximum. We used the default settings and run only the CPU test. Each test was run thrice.

Far Cry and Doom 3: Both these games boast high graphic details, and are quite taxing on the processor, making them ideal processor benchmarks. We ran both of them at 640 x 480 and 800 x 600 to put more stress on the CPU.

Company of Heroes: This is the latest in the breed of RTS games and one that can bring the best of machines down to their knees with its ultra. Settings were kept at high, and since the game doesn't allow 640 x 480, we used 800 x 600 and 1024 x 768. The built-in benchmark gives the resulting fps.

DivX Encoding: We encoded a 100 MB VOB file to DivX using the DivX 6.2.5 video converter. This benchmark is run thrice for each processor, and the final score is an average of the three encoding times.

CINEBENCH 10.1: CINEBENCH 10.1 is a free benchmarking tool for Windows and Mac OS. It is based on the CINEMA 4D engine. CINEBENCH 10 tests processors for OpenGL processing, multithreading and multi-processing.

POV-Ray 3.6: The Persistence Of Vision raytracer, a tool that creates an image by mathematical calculation. The number-crunching involved is processor-intensive, making it a great processor benchmark. In the default installation, we used the "Chess2" and "landscape" scenes to be rendered at 1024 x 768 resolution without anti-aliasing.

ScienceMark 2.0: ScienceMark 2.0, as the name suggests, is a benchmark based on scientific calculations. This benchmark consists of multiple benchmarks that test the various aspects of a processor. We use the Molecular dynamics, Primordia, Cryptography, Stream and Memory benchmarks.

WinRAR 3.7: WinRAR 3.7 offers a built-in benchmark, operating on random data that is processed in the memory—this prevents hard disk performance from affecting the test results.

W Prime: A small freeware utility that calculates square-roots of large of numbers. This utility is multi-threaded, so takes good advantage of multi-core processors. We use the 32 M calculations option.

SiSoft Sandra 2007: This is the good old benchmarking tool with newer tests. We logged scores particularly related to the CPU, such as Dryhstone, Whetstone, Multimedia index (Integer and Float) and Cache performance for a 4 MB data block.